

Tianning Feng

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SUMMARY

UPenn CIS Master's student focused on AI agent applications, full-stack engineering, and machine learning systems. Turns product problems and cross-product capabilities into executable agent workflows, carrying them through launch, validation, and iteration.

EDUCATION

University of Pennsylvania, MSE in Computer & Information Science 2025.08 – 2027.05

- Relevant Coursework: Artificial Intelligence, Database Systems, Embedded Systems.

Emory University, B.S. in Computer Science & B.S. in Psychology · GPA 3.794/4.0 2021.08 – 2025.05

- Honors: Dean's List 2024 Fall-2025 Spring, Dean's List 2024 Spring-2024 Fall.
- Relevant Coursework: Data Mining, Machine Learning, Data Structures & Algorithms, Cognitive Psychology.

PROFESSIONAL EXPERIENCE

ByteDance · Data Platform (Shanghai) – Agent Application Engineer Intern 2026.01 – Present

- End-to-End Data Workflow Skills: aimed to keep the full development loop in one product; contributed to dataset, experiment, and service Skills that orchestrate cross-product capabilities and validate before execution, growing successful dataset Skill calls 520% in week two.
- Cross-System Changelog Pipeline: identified a user-facing product information gap and independently delivered the 0-to-1 path from code changes and AI drafts through human review to cross-repository display, then rebuilt repetitive records as weekly snapshots.
- Skill Runtime Controls: co-owned design through launch across teams, enabling users to switch and compare versions while unified resolution and runtime checks give the agent only necessary, authorized Skills and prevent disabled capabilities from being invoked.

Liba Space (San Jose, CA) – Full-stack Engineer Intern 2025.09 – 2025.12

- Owned and led development of the Resume Analysis & Rewrite feature from 0-1 for Jobnova, a North American job platform; seamlessly integrated it into the user flow, leading to a 50% increase in MAU.
- Collaborated with cross-functional teams to translate user needs into technical specifications and validation logic, leading the full development lifecycle from initial coding to production deployment.
- Optimized backend API workflows to reduce LLM token consumption and improve response quality; refined the system architecture based on user feedback and managed final quality assurance (QA).

Government-Sponsored AI Education Initiative (Beijing, China) – Contract Product Leader 2025.05 – 2025.08

- Led the technical execution of a 0-1 RAG platform, managing the full development lifecycle from database schema design (SQL/Vector) to API integration for automated lesson planning.
- Enhanced system response quality by refactoring the retrieval layer with a hybrid BM25 + Vector approach, significantly boosting the relevance of retrieved academic contexts.
- Automated content generation workflows by building a parsing engine that transformed raw educational PDFs into structured formats, reducing manual PPT preparation time by 70%.

Shen Tong Technology Group Co., Ltd (Ningbo, China) – Product Intern 2021.06 – 2021.08

- Improved usability scores by approximately 30% by analyzing user interaction logs and delivering technical specifications to resolve dashboard navigation bottlenecks.
- Accelerated design sign-off by 20% by establishing a standardized feedback loop between Design and Engineering stakeholders, effectively reducing cross-functional communication overhead.

RESEARCH EXPERIENCE

Melody Lab · Emory University (Atlanta, GA) – Deep Learning Researcher 2023.09 – 2025.05

- Developed SeizureFormer, a novel Transformer-based time-series forecasting model in PyTorch to predict seizure risks using RNS data, achieving state-of-the-art performance in 1-14 day horizons.
- Engineered a Dynamic Hypergraph Neural Network (EpiDHGNN) for epidemic modeling, optimizing the message-passing algorithm to achieve 30% faster convergence and 7-12% higher accuracy than baselines.
- Authored 2 papers accepted at PSB 2026; SeizureFormer was selected for an oral presentation (Top 10%).

Language Biomarker Lab · Emory University (Atlanta, GA) – Research Assistant 2022.09 – 2024.11

- Reduced manual data processing workload by 80% by architecting automated ETL pipelines using Hugging Face and PyTorch for multi-modal data.
- Standardized 100+ GB of raw data into structured formats for downstream ML tasks, implementing efficient data loading strategies to optimize training throughput.

PROJECT

PhillyHome Insight — Housing & Community Analytics Platform 2025.05 – Present

- Gained 200+ early subscriptions by leading the full-stack engineering of an analytics platform that integrates heterogeneous datasets (property, census, crime) via SQL.
- Achieved sub-200ms query latency for complex geospatial queries by designing composite database indexes and optimizing RESTful API response structures.

OpenAPS Simulation — OpenAPS Simulation 2022.09 – 2024.11

- Achieved sub-100ms response latency in a safety-critical environment by optimizing C++ task scheduling and memory management on FreeRTOS.
- Engineered a reliable PID control loop to regulate insulin delivery based on real-time CGM sensor data, simulating medical-grade device standards.

Smart Home Assistant Development — Smart home assistant 2024.01 – 2025.01

- Enabled concurrent control of 15+ smart devices by architecting a scalable backend using MongoDB and Swift with real-time synchronization.
- Reduced data sync latency to under 200ms by implementing real-time database observers (Change Streams) and optimizing the data transmission layer.

SKILL

- **Programming:** Python, C/C++, Java, JavaScript/TypeScript (React, Node.js), SQL, AI Agentic Programming
- **Machine Learning:** PyTorch, Hugging Face, Transformers, Graph Neural Networks, Reinforcement Learning
- **Infrastructure & Tools:** Docker, Git, AWS, MongoDB, Neo4j, Vector Databases, FreeRTOS, RESTful API
- **Languages:** Chinese (Native), English (Fluent)

PUBLICATION

- Feng, T., Ni, J., Gleichgerricht, E., Jin, W. SeizureFormer: A Transformer Model for IEA-Based Seizure Risk Forecasting. Pacific Symposium on Biocomputing (PSB 2026) · Oral presentation · Hawaii, 2026.
- Liu, S., Gong, S., Feng, T., Liu, Z., Lau, M. S. Y., Jin, W. Higher-order Interaction Matters: Dynamic Hypergraph Neural Networks for Epidemic Modeling (EpiDHGNN). Pacific Symposium on Biocomputing (PSB 2026).